

PRANITA KHANAL

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Education

University of Texas at Arlington

Bachelor of Science in Computer Science, Minor in Mathematics

Arlington, TX

Aug 2023 – May 2027

Work Experience

Department of Physics

Lab Technician

The University of Texas at Arlington

Jan 2024 – Aug 2026

- Modified lab manuals, create new experiments, and manage inventory logistics by conducting vendor cost-benefit analyses to optimize the departmental budget.
- Managed the installation and maintenance of lab computers and digital interfaces; lead the integration of PASCO Capstone and Data Studio software to modernize data acquisition and streamline experiment workflows.
- Utilized AutoCAD to engineer custom mechanical components for Mechanics labs; perform hardware repairs, including precision soldering and circuitry to maintain assets.

CyberGuard Lab

Research Assistant

The University of Texas at Arlington

May 2026 – Present

- Set up and debugged a full VLM research environment on an HPC cluster, enabling GPU-accelerated evaluation of multiple VLMs across multiple visual QA benchmarks (TextVQA, POPE, MMBench)
- Developed an adversarial attack pipeline against PruMerge token compression, showing perturbed images bypass the compression mechanism's stride cliff, increasing token count by 5 times.

Projects

RAG PDF Chatbot

Python, CSS, HTML, JS

Dec 2025

[pdf-chatbot-frontend-steel.vercel.app](#)

- Built a full-stack AI chatbot with FastAPI backend and JS frontend using RAG architecture with Pinecone vector database and Google Gemini 2.5 API.
- Deployed containerized microservices on Hugging Face Spaces and Vercel with a CI/CD pipeline via GitHub for automated deployments.
- Engineered semantic search using Sentence Transformers for local embeddings, optimizing API costs while maintaining response accuracy.

Neural Network Library

Python, NumPy

Mar 2026

[NNL@github](#)

- Built a modular neural network library from scratch supporting arbitrary depth/width with linear layers, ReLU, sigmoid, and full backpropagation.
- Applied library to NYC taxi trip duration prediction, performing feature engineering, normalization, and hyperparameter search across 3+ model configurations.
- Validated XOR classification with a custom 2-layer network, comparing sigmoid vs. tanh activations to analyze convergence behavior.

Technical Skills

Languages: Python, Java, C, JavaScript, C++, SQL, PHP, Assembly, R, HTML/CSS

Frameworks & Libraries: Spring Boot, Spring Data JPA, Apache Kafka, FastAPI, React.js, RESTful API

AI/ML Tools: Scikit-learn, Pandas, NumPy, Matplotlib, Sentence Transformer

Data Science & Analytics: Data Analysis, Data Modeling, Data Visualization

DevOps & Tools: Docker, Git/GitHub, CI/CD Pipeline, Maven, Linux/UNIX, API Integration

Leadership / Extracurricular

Vice President & Treasurer

Nepalese Students' Association

Summer, Fall 2026

University of Texas at Arlington

- Represent and advocate for 300+ Nepali students at UTA, organizing 10+ cultural events and managing chapter finances to support community engagement and organizational growth.

Project Instructor

UK SARAID, The Japan Foundation

2018 – 2020

Kathmandu, Nepal

- Coordinated SMART Camp and instructed the HANDS-selected CAC project "SMART 2019" for the Kathmandu district with 150 participants; assisted in SMART's international COVID-19 session with cross-country research teams.